AP Physics 1

Name _____

Period _____ Date _____

Practice Test – Introductory Dynamics (Version 6)

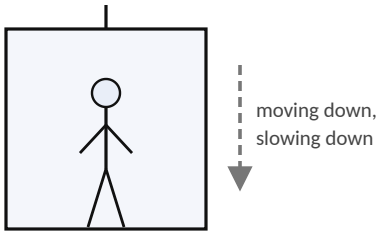
On any problems, you must show all work, include units, round properly, and box your final answer. Do the algebra first, before plugging in values.

Rewrite the following statement in the space below: On my honor, before God, I have neither given, nor received, any aid on this test.

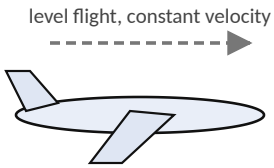
For all questions:

- Use g = 9.8 m/s² (9.8 N/kg). Air resistance is negligible unless the question says it is not.
- Free-body diagrams: draw the forces themselves, not components, each as an arrow starting on the dot.
- Round answers to the significant figures of the given values.
- 50 minutes. Four questions, 25 points each (100 points total).

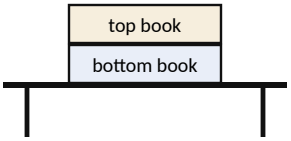
1. (25 points) For each situation, draw and label the forces (not components) exerted on the object named. Draw each force as its own arrow starting on the dot and pointing away from it, with lengths that show which forces are larger.



(a)



(b)



(c)

(a) A person stands in an elevator that is moving down and slowing down. Draw the free-body diagram of the person.

(b) An airplane flies to the right in level flight at constant velocity. Draw the free-body diagram of the airplane.

(c) Two identical books are stacked at rest on a table. Draw the free-body diagram of the **bottom** book.

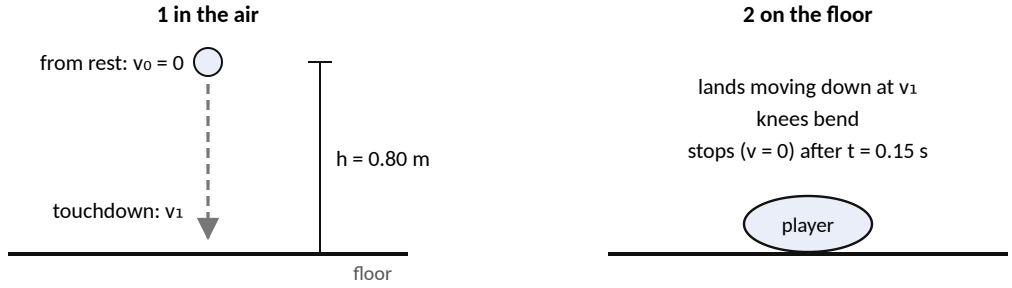
(a) person

(b) airplane

(c) bottom book

(d) In part (c), is the normal force from the table on the bottom book equal to the weight of the bottom book? Explain.

2. (25 points) A 70 kg basketball player drops from rest from a height of 0.80 m above the floor. After landing, the player bends their knees and comes to a stop in 0.15 s. Assume the floor's push is constant while the player stops.



(a) Calculate the player's speed just as their feet touch the floor.

(b) Calculate the player's average acceleration while stopping, magnitude and direction.

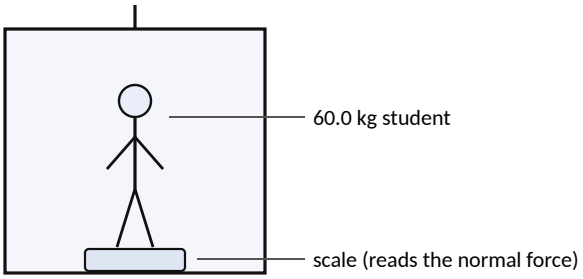
(c) Draw the free-body diagram of the player while stopping. Then calculate the normal force from the floor on the player.

(c) player, stopping

(d) How many times the player's weight is this normal force? What force does the player exert on the floor (magnitude and direction)?

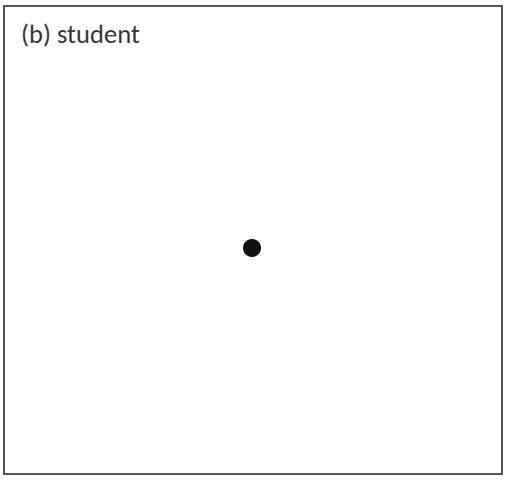
(e) In one sentence, explain why landing with bent knees reduces the force on the player's legs.

3. (25 points) A 60.0 kg student stands on a bathroom scale in an elevator. The scale reads the size of the normal force it exerts on the student.



(a) The elevator is at rest. What does the scale read?

(b) The scale reads 660 N. Draw the free-body diagram of the student, then calculate the student's acceleration, magnitude and direction. Describe **both** ways the elevator could be moving.



(c) The scale reads 520 N. Calculate the student's acceleration, magnitude and direction. Describe **both** ways the elevator could be moving.

(d) If the elevator cable broke and the elevator were in free fall, what would the scale read? Explain using Newton's second law.

4. (25 points) Students drop stacks of nested coffee filters and measure each stack's terminal speed v_t. Adding filters changes the mass m but not the shape. At terminal speed the stack falls at constant velocity, so the drag force equals the weight. For these filters the drag force is $F_D = kv_t^2$, so $mg = kv_t^2$, where k is a constant.

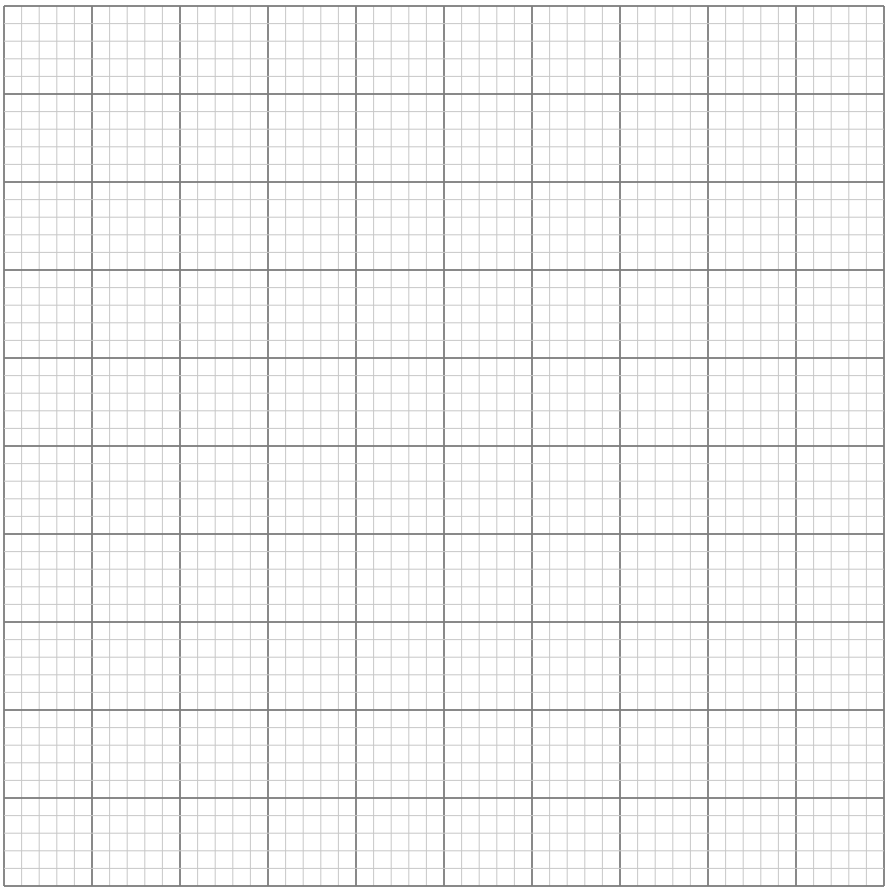


The students graph the weight $F_g = mg$ on the vertical axis and v_t^2 on the horizontal axis.

Mass m (g)	Terminal speed v_t (m/s)	F_g (N)	v_t^2 (m^2/s^2)
1.0	1.81		
2.0	2.55		
3.0	3.07		
4.0	3.59		
5.0	4.09		
6.0	4.47		

(a) Calculate F_g and v_t^2 for each trial and fill in the table. Show one sample calculation of each below.

(b) Plot F_g versus v_t^2 on the grid. Label each axis with the quantity and units, and choose a scale.



(c) Draw the best-fit line. Use two far-apart points on your line to calculate its slope, with units. The slope is k.

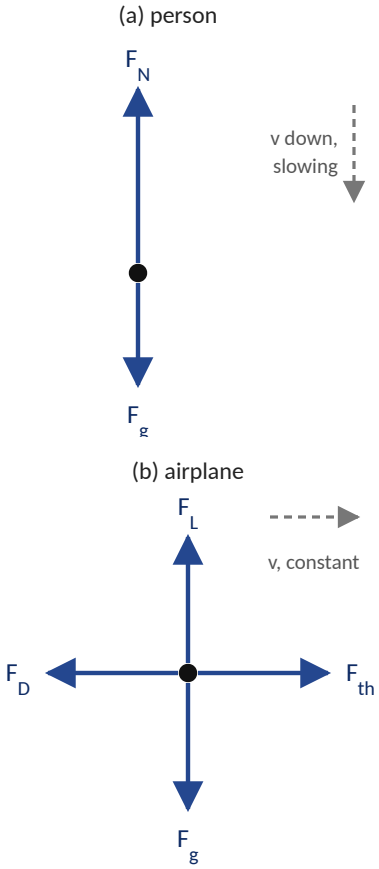
(d) Use your value of k to predict the terminal speed of a stack with a mass of 10.0 g.

Answer Key – Practice Test, Version 6 – Introductory Dynamics

Worked step by step, the same way as the class notes: sketch, free-body diagram, $a = F_{net}/m$, forces with signs, algebra first, then numbers with units. $g = 9.8\ m/s^2$; answers match the significant figures of the givens.

Every problem: $a = \Sigma F/m$ (equation sheet), written $a = F_{net}/m$. Replace F_{net} with the actual forces and their signs ($F_N - F_g$, $F_{th} - F_D$, $F_L - F_g$), swap $F_g = mg$, solve in letters, then plug in.
Kinematics, the first three on the sheet: #1 $v = v_0 + at$ · #2 $\Delta x = v_0 t + \frac{1}{2}at^2$ · #3 $v^2 = v_0^2 + 2a\Delta x$. Pick the one that leaves out the variable you don't know and don't need.

1. Free-body diagrams (25 points)



Step 1 **List every force on the person.**

Gravity F_g pulls down. The elevator floor pushes up (F_N). Nothing else touches the person.

Step 2 **Check the lengths.**

Moving down and slowing down means the acceleration points up. $a = F_{net}/m$, so F_{net} points up too: $F_N > F_g$. The F_N arrow is longer.

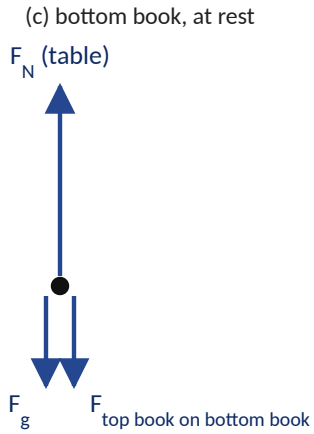
Watch out: the velocity is down but the acceleration is up. Arrow lengths follow the acceleration, not the motion.

Step 3 **Constant velocity means $a = 0$, so $F_{net} = 0$ both ways.**

Vertical: lift F_L up balances gravity F_g down, so $F_L = F_g$ (equal arrows).

Horizontal: thrust F_{th} forward balances drag F_D backward, so $F_{th} = F_D$ (equal arrows).

Watch out: no extra arrow to the right for the motion. Thrust only has to cancel drag; the plane keeps moving because of inertia.



Step 4 **Only the forces on the bottom book.**

Its own weight F_g pulls down. The top book pushes down on it ($F_{\text{top book on bottom book}}$). The table pushes up ($F_N$). At rest, so $F_{net} = 0$: $F_N = F_g + F_{\text{top book on bottom book}}$. The up arrow is as long as the two down arrows together.

Watch out: the top book's weight acts on the top book, so it does not go on this diagram. What reaches the bottom book is the push of the top book on it.

Step 5 **(d) Is F_N equal to the bottom book's weight?**

The top book is at rest, so the bottom book pushes up on it with a force equal to the top book's weight. Third law: the top book pushes down on the bottom book with the same size, so $F_{\text{top book on bottom book}}$ = weight of the top book.

Bottom book at rest: $F_N = F_g + F_{\text{top book on bottom book}}$ = (weight of bottom book) + (weight of top book).

No. F_N equals the weight of both books, twice the bottom book's weight

2. Basketball landing: two stages (25 points)

Step 1 **Two accelerations, so two pictures. Up is +.**

stage	known	wanted
1 in the air	$v_0 = 0$, $a = -g$, $\Delta y = -h = -0.80$ m	v_1 at touchdown
2 on the floor	v_1 down, $v = 0$, $t = 0.15$ s, $m = 70$ kg	a_2, then F_N

Step 2 **(a) In the air: time is not given or asked, so use #3.**

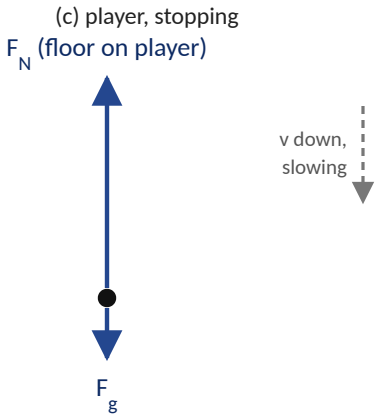
$v_1^2 = v_0^2 + 2a\Delta y \rightarrow v_1^2 = 0 + 2(-g)(-h) \rightarrow v_1 = \sqrt{2gh}$

$v_1 = \sqrt{2(9.8\ \text{m/s}^2)(0.80\ \text{m})} = \sqrt{15.68\ \text{m}^2/\text{s}^2} = 3.96$ m/s → **4.0 m/s, downward** (keep 3.96 m/s)

Step 3 **(b) On the floor: acceleration from the change in velocity.**

$$a_2 = \frac{\Delta v}{\Delta t} = \frac{0 - (-v_1)}{t} = \frac{v_1}{t} = \frac{3.96\ \text{m/s}}{0.15\ \text{s}} = 26.4\ \text{m/s}^2 \rightarrow$$ **26 m/s², upward**

Moving down and slowing down, so the acceleration points up. The + sign agrees.



The acceleration is up, so the net force is up: the F_N arrow is much longer than F_g (about 3.7 times).

Step 4 **(c) Newton's second law while stopping.**

$a = \frac{F_{net}}{m} \rightarrow a_2 = \frac{F_N - F_g}{m} \rightarrow F_N - mg = ma_2 \rightarrow F_N = m(g + a_2)$

$F_N = (70\text{ kg})(9.8\text{ m/s}^2 + 26.4\text{ m/s}^2) = (70\text{ kg})(36.2\text{ m/s}^2) = 2534\text{ N}$

$F_N = 2.5 \times 10^3$ N, upward (two significant figures, matching 70 kg, 0.80 m and 0.15 s)

Step 5 **(d) Compare with the weight.**

$\frac{F_N}{F_g} = \frac{m(g + a_2)}{mg} = 1 + \frac{a_2}{g} = 1 + \frac{26.4\text{ m/s}^2}{9.8\text{ m/s}^2} = 3.69 \rightarrow$ **3.7 times the player's weight**

Check: $F_g = mg = (70\text{ kg})(9.8\text{ m/s}^2) = 686$ N, and 2534 N ÷ 686 N = 3.7.

Step 6 **(d) Third law last: the question asks for the force on the floor.**

$F_{\text{player on floor}} = F_{\text{floor on player}} = F_N$, opposite direction. **$2.5 \times 10^3$ N, downward on the floor**

Step 7 **(e) Bent knees.**

Bending the knees makes the stopping time longer, so $a = \Delta v/t$ is smaller and $F_N = m(g + a)$ is smaller.

Watch out: the 0.15 s is only the stopping time on the floor. Never use it for the fall, and never use the 0.80 m for the stop.

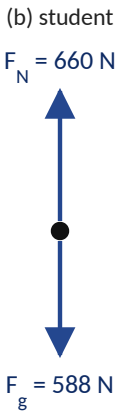
3. Elevator scale (25 points)

Step 1 **The scale reads F_N. Up is +. Solve in letters once.**

$a = \frac{F_{net}}{m} \rightarrow a = \frac{F_N - F_g}{m} \rightarrow F_N - mg = ma \rightarrow F_N = m(g + a)$, so $a = \frac{F_N}{m} - g$

Step 2 **(a) At rest: a = 0.**

$F_N = m(g + 0) = mg = (60.0\text{ kg})(9.8\text{ m/s}^2) =$ **588 N**



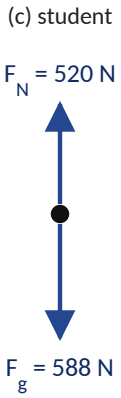
Step 3 **(b) Scale reads 660 N, more than the weight.**

$a = \frac{660\text{ N}}{60.0\text{ kg}} - 9.8\text{ m/s}^2 = 11.0\text{ m/s}^2 - 9.8\text{ m/s}^2 = 1.2\text{ m/s}^2 \rightarrow$

1.2 m/s², upward

(1 N/kg = 1 m/s².) The F_N arrow is longer, so the net force and the acceleration point up.

Moving up and speeding up, or moving down and slowing down



Step 4 **(c) Scale reads 520 N, less than the weight.**

$$a = \frac{520\ \text{N}}{60.0\ \text{kg}} - 9.8\ \text{m/s}^2 = 8.67\ \text{m/s}^2 - 9.8\ \text{m/s}^2 = -1.13\ \text{m/s}^2 \rightarrow$$

1.1 m/s², downward

Moving down and speeding up, or moving up and slowing down

Watch out: the scale reading gives the direction of the acceleration, not the direction of the velocity. That is why two motions fit each reading.

Step 5 **(d) Free fall: a = −g.**

$F_N = m(g + a) = m(g + (-g)) = m(0) =$ **0 N**

Gravity alone already gives the student an acceleration of g downward, so the scale does not have to push at all. The student and the scale fall together.

4. Linearize the data (25 points)

Step 1 **Match $mg = kv_t^2$ to $y = mx + b$.**

$F_g = k \cdot v_t^2 + 0$

vertical (y)	horizontal (x)	slope	intercept
F_g (N)	v_t^2 (m²/s²)	k	0

Step 2 **(a) Grams to kilograms first, then fill both columns.**

$m = 1.0\ \text{g} = 0.0010\ \text{kg} \rightarrow F_g = mg = (0.0010\ \text{kg})(9.8\ \text{m/s}^2) = 0.0098\ \text{N}$

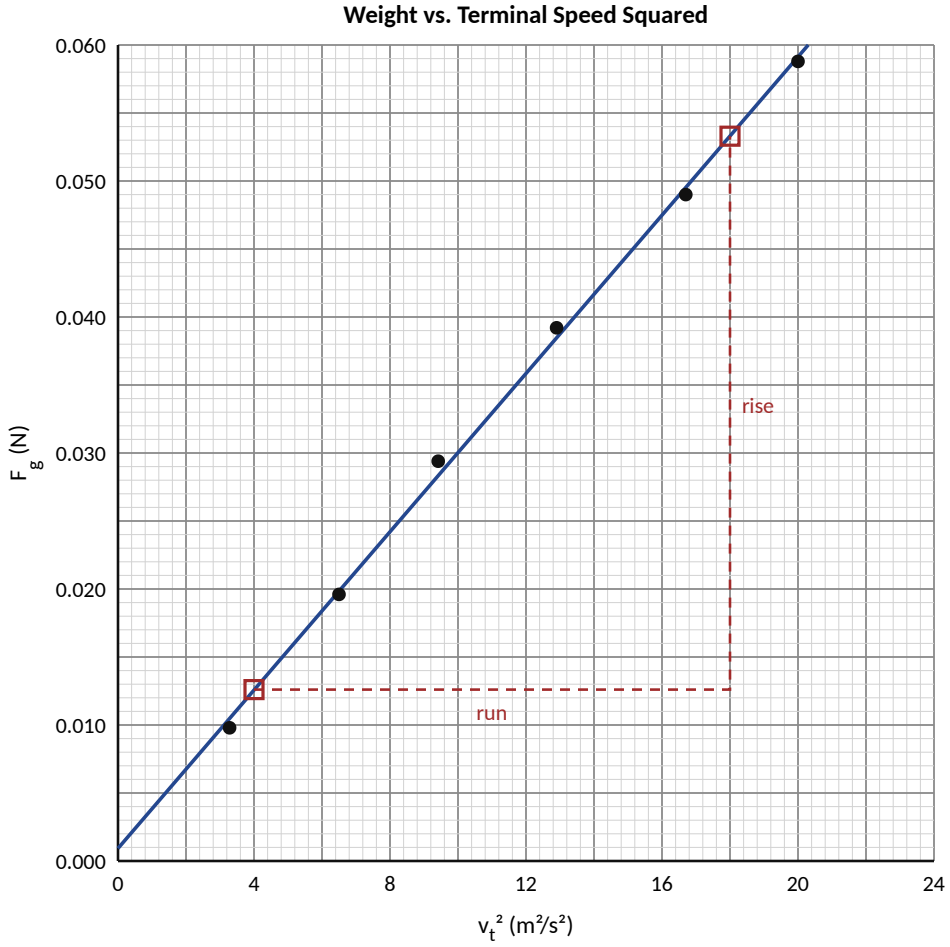
$v_t^2 = (1.81\ \text{m/s})^2 = 3.28\ \text{m}^2/\text{s}^2$

m (g)	v_t (m/s)	F_g (N)	v_t^2 (m²/s²)
1.0	1.81	0.0098	3.28
2.0	2.55	0.0196	6.50
3.0	3.07	0.0294	9.42
4.0	3.59	0.0392	12.9
5.0	4.09	0.0490	16.7
6.0	4.47	0.0588	20.0

0.020 N in place of 0.0196 N (and so on) also earns credit.

Watch out: using grams makes every weight 1000 times too big. 1 g = 0.001 kg.

Step 3 **(b) and (c) Graph: labeled axes with units, a scale that uses more than half the grid, one straight line through the middle of the points.**



The line is not forced through the origin. The red squares are points read off the line, not data points.

Step 4 **(c) Slope from two far-apart points on the line, units carried.**

Points on the line: (4.0 m²/s², 0.0126 N) and (18.0 m²/s², 0.0533 N)

$$\text{slope} = \frac{0.0533\ \text{N} - 0.0126\ \text{N}}{18.0\ \text{m}^2/\text{s}^2 - 4.0\ \text{m}^2/\text{s}^2} = \frac{0.0407\ \text{N}}{14.0\ \text{m}^2/\text{s}^2} = 0.00291\ \text{N}\cdot\text{s}^2/\text{m}^2$$

Units: $\text{N}\cdot\text{s}^2/\text{m}^2 = (\text{kg}\cdot\text{m}/\text{s}^2)(\text{s}^2/\text{m}^2) = \text{kg/m}$. **$k \approx 2.9 \times 10^{-3}$ kg/m**

Any value from 2.7×10^{-3} kg/m to 3.1×10^{-3} kg/m from a reasonable line earns credit.

Step 5 **(d) Predict: solve $mg = kv_t^2$ for v_t in letters, then plug in.**

$$mg = kv_t^2 \rightarrow v_t = \sqrt{\frac{mg}{k}}$$

$$v_t = \sqrt{\frac{(0.0100\ \text{kg})(9.8\ \text{m/s}^2)}{0.00291\ \text{kg/m}}} = \sqrt{33.7\ \text{m}^2/\text{s}^2} = 5.81\ \text{m/s} \rightarrow \mathbf{v_t \approx 5.8\ m/s}$$

Any value from 5.6 m/s to 6.0 m/s that follows from the student's own slope earns credit.

Watch out: 10.0 g is 0.0100 kg, and the graph gives v_t^2, so take the square root at the end.

1. Free-body diagrams

Suggested partial credit	pts
(a) Normal force up and gravity down, each starting on the dot, labeled	3
(a) F_N longer than F_g (acceleration up)	3
(b) Thrust right, drag left, lift up, gravity down	4
(b) Thrust equals drag and lift equals gravity (equal pairs)	3
(c) F_N up, F_g down, force of top book on bottom book down; no top-book weight	4
(c) F_N as long as the two down arrows combined	3
(d) No: it equals the weight of both books	2
(d) Explanation: $F_{net} = 0$ on the bottom book, and the top book pushes down with its own weight	3
Total	**25**

2. Basketball landing

Suggested partial credit	pts
(a) $v = \sqrt{(2gh)}$ from #3, letters first	3
(a) 4.0 m/s with units	2
(b) $a = \Delta v/t$ using the 0.15 s stopping time	3
(b) 26 m/s², upward	2
(c) Normal force up, much longer than gravity down; labeled	3
(c) $F_N - mg = ma$ solved for F_N in letters	3
(c) 2.5×10^3 N, units in every parenthesis	2
(d) 3.7 times the weight	2
(d) Third law: player on floor, 2.5×10^3 N, downward	2
(e) Longer stopping time, so smaller a and smaller F_N	3
Total	**25**

3. Elevator scale

Suggested partial credit	pts
(a) $F_N = mg = 588$ N	3
(b) Diagram: F_N up longer than F_g down	2
(b) $F_N - mg = ma$ solved for a in letters	3
(b) 1.2 m/s², upward	2
(b) Both motions: up and speeding up, or down and slowing down	3
(c) 1.1 m/s², downward	3
(c) Both motions: down and speeding up, or up and slowing down	3
(d) $a = -g$ substituted into $F_N = m(g + a)$	3
(d) 0 N; gravity alone gives the acceleration, so the scale does not push	3
Total	**25**

4. Linearize the data

Suggested partial credit	pts
(a) Grams converted to kilograms	3
(a) F_g column correct	2
(a) v_t^2 column correct	2
(b) Axes labeled with quantity and units (F_g vertical, v_t^2 horizontal)	2
(b) Scale uses more than half the grid	2
(b) Points plotted correctly	2
(c) One straight best-fit line, not forced through the origin	2
(c) Two far-apart points read off the line, not data points	2
(c) k between 2.7×10^{-3} kg/m and 3.1×10^{-3} kg/m, with units	3
(d) $v_t = \sqrt{(mg/k)}$ in letters first	2
(d) 0.0100 kg used; 5.6 m/s to 6.0 m/s	3
Total	**25**